SCIS & ISIS 2026

# Comparing Problem-Specific and Mean Hidden-State Interventions for Context-Reconstructed Reasoning

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Slides (PDF)

# LLMs can fail after a context change



**Original (ORG)**

A man fell from a 100-rung ladder, but was unhurt. How?






Correct answer: the lowest rung

**Context-Reconstructed (CR)**

A man fell from a 50-story building, but was unhurt. How?






Correct answer: the lowest floor

**Different contexts, the same underlying solution structure.**

Illustrative example. Schematic responses, not measured.

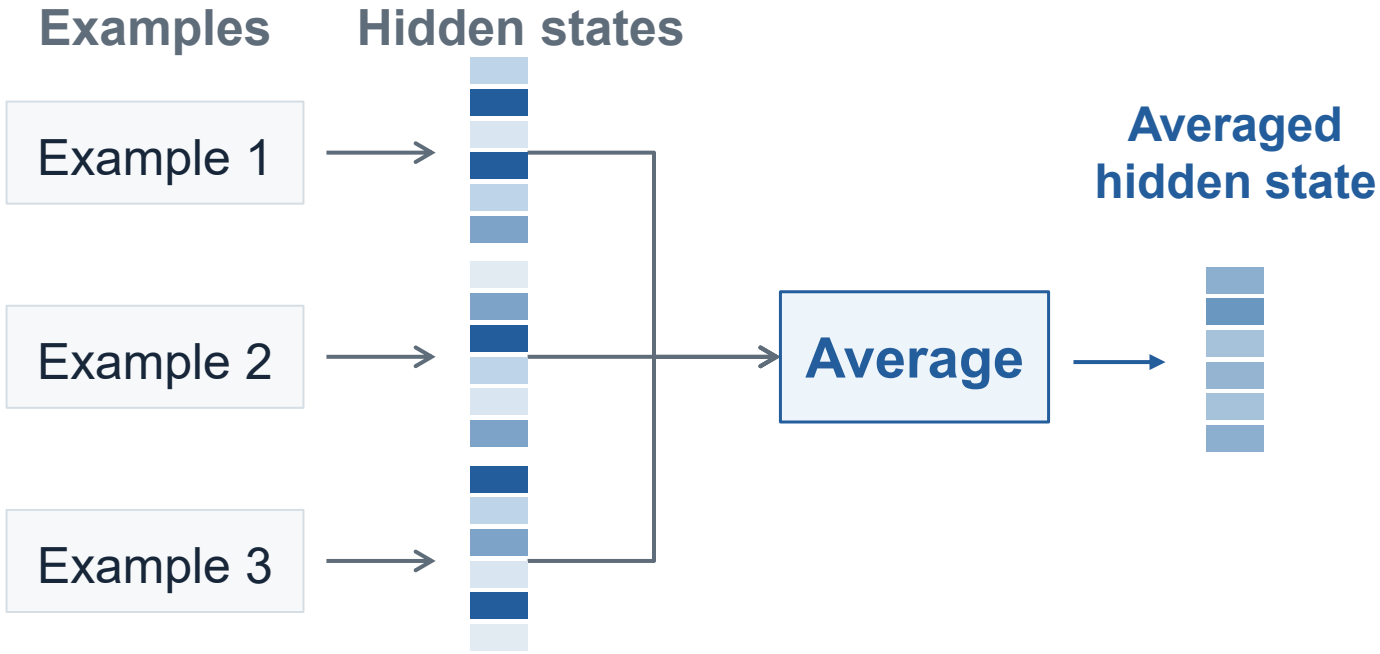
# Related Work: Interventions with averaged states



## Hidden states

Numerical vectors computed at each token and layer

- Changing hidden states can alter model behavior. [6–8]
- Many interventions use average directions. [6–8]




### Potential limitation

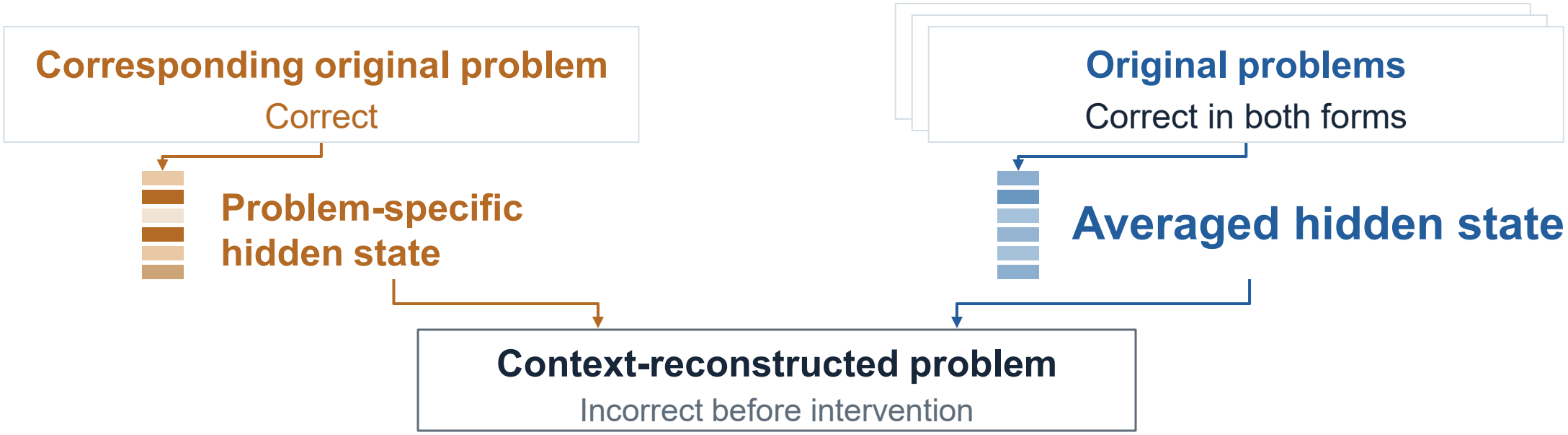
**Averaging may dilute problem-specific information needed to recover a reasoning failure.**

Li et al. (2023), Liu et al. (2024), Todd et al. (2024)

# Research Question



## Does a problem-specific hidden state recover reasoning failures more effectively than an averaged hidden state?




**Main finding**

## Problem-specific states give higher peaks in all four models

# Problem pairs and answer evaluation



**BrainTeaser Sentence Puzzle (BTSP)** — **209 problem pairs**

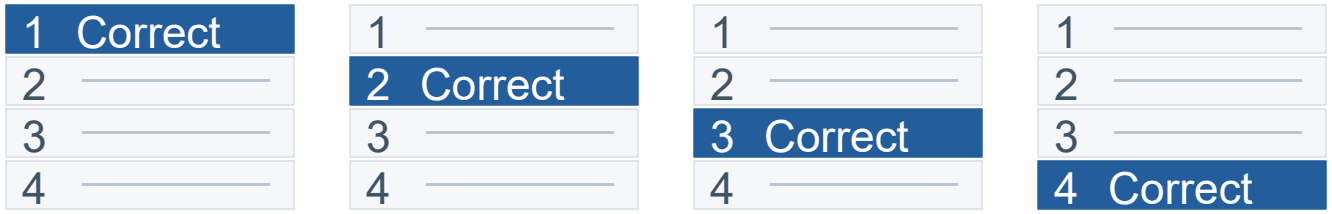
Four-choice lateral-thinking word puzzles

**Original (ORG)** → **Context-Reconstructed (CR)**

| Input | Each model | Output |
|---|---|---|
| Question + four answer options → | • Gemma 2 9B IT • Gemma 3 12B IT<br>• Gemma 4 E2B IT • Gemma 4 E4B IT | → Answer number 1–4 |

Zero-shot. Select the highest candidate log-likelihood.

Same correct option, four positions (ORG / CR separately)




Answer 1 ✓ Answer 2 ✓ Answer 3 ✓ Answer 4 ✓

→ All 4 answers correct

**Problem correct**

Any error: incorrect

# Failures after a context change



Letter order: ORG / CR  C = Correct  I = Incorrect

## Targets for recovery

CI group = Correct / Incorrect

## Sources for averaging

CC group = Correct / Correct

Problem pairs, classified separately for each model

| Model | CI group | CC group |
|---|---|---|
| Gemma 2 9B IT | **40** | 113 |
| Gemma 3 12B IT | **38** | 94 |
| Gemma 4 E2B IT | **35** | 11 |
| Gemma 4 E4B IT | **32** | 99 |

# Choosing hidden states



**Problem-specific (Pair-CI)**

Hidden state from the corresponding ORG problem

**Averaged (Mean-CC)**

Average of ORG hidden states from the CC group

| Additional comparisons | ORG source | Comparison purpose |
|---|---|---|
| **Mean-CI** | Averaged ORG hidden state from other CI problems | Source-group membership |
| **Surface-similar-CI / Surface-similar-CC** | ORG with highest TF-IDF similarity to CR | Lexical similarity |
| **Random-CI / Random-CC** | One randomly selected ORG | Arbitrary source choice |

CI-source controls use Gemma 4 models and exclude the corresponding ORG problem.

# Hidden-state intervention

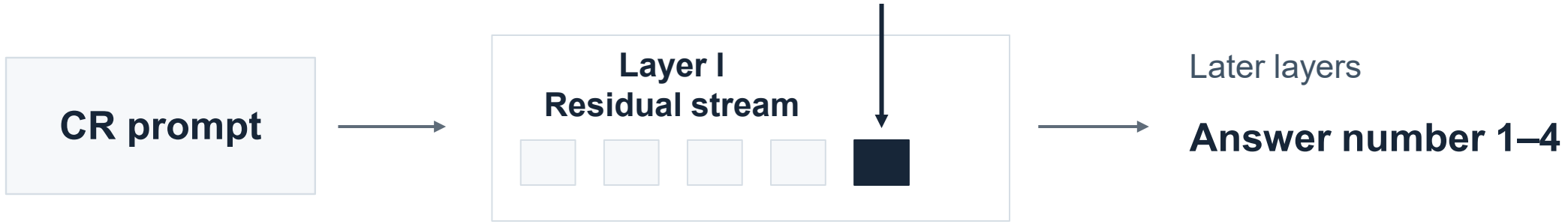


**Source hidden state**

Average hidden states over four option-slot variants per ORG

**Norm-matched source**

Match the norm of the target problem's hidden state




$$\tilde{h}_{CR} = h_{CR} + \alpha(\hat{h}_{src} - h_{CR})$$

$h_{CR}$: target state $\hat{h}_{src}$: norm-matched source

- α = 1.0

  Replace with the norm-matched source

- Final prompt position
- All layers, one at a time

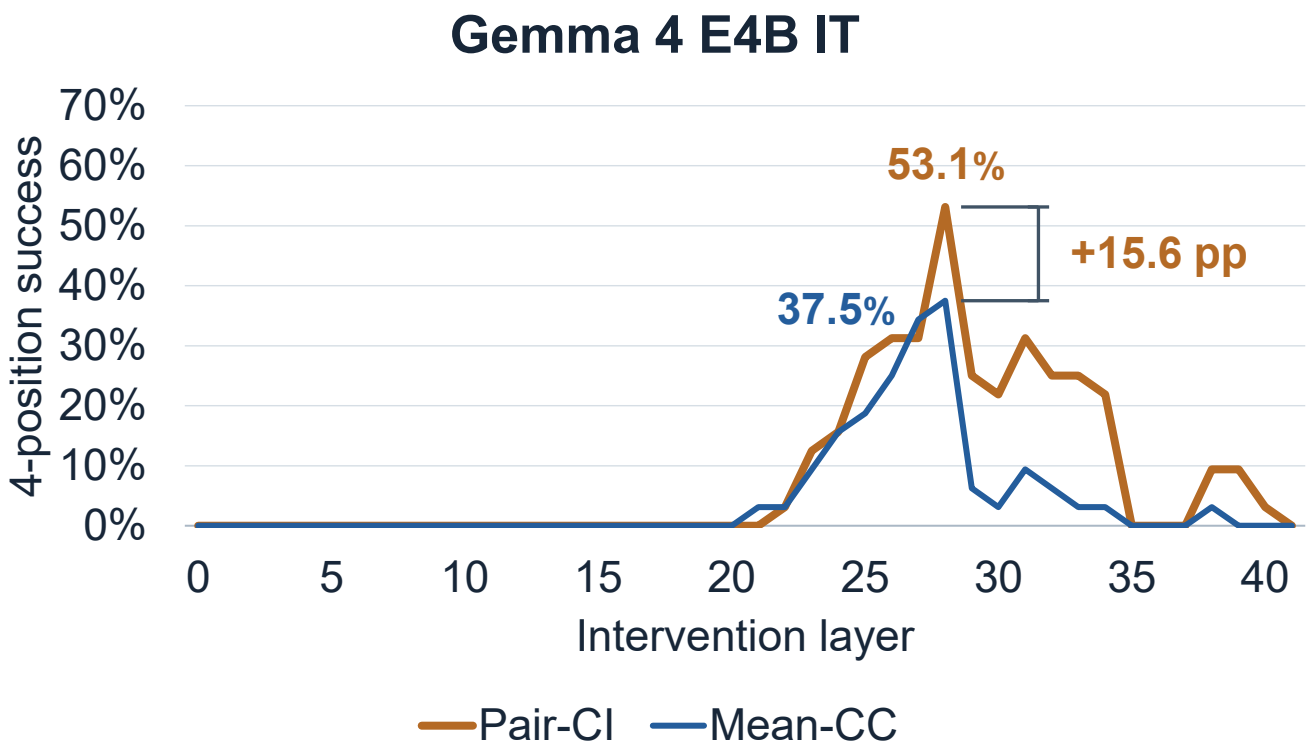
**4-position success rate**

CI group CR targets correct in all four variants (%)

# Higher peak recovery in all four models



**α = 1.0** Gaps in percentage points (pp)




**Peak recovery**

| Model | Pair-CI | Mean-CC |
|---|---|---|
| Gemma 2 9B IT | **22.5%** | 15.0% |
| Gemma 3 12B IT | **15.8%** | 7.9% |
| Gemma 4 E2B IT | **65.7%** | 28.6% |
| Gemma 4 E4B IT | **53.1%** | 37.5% |

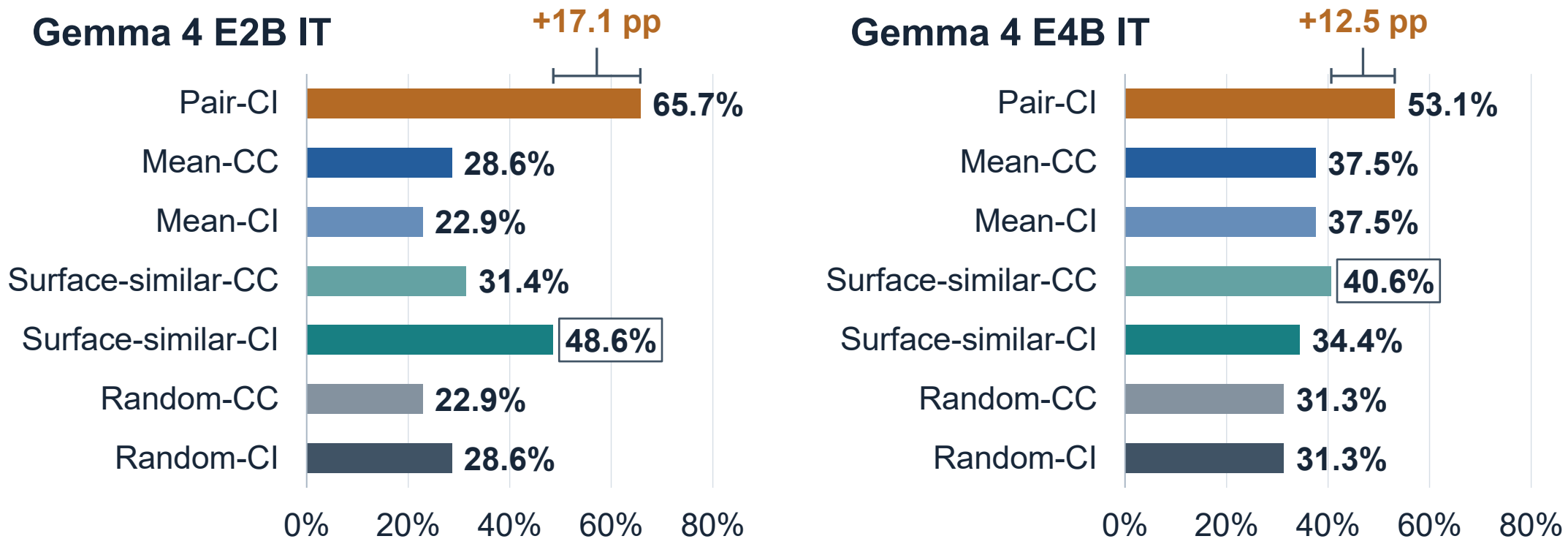
Peak over layers

**Averaging across problems may weaken useful information.**

# Other sources give lower peaks



Gaps: Pair-CI vs best alternative; peaks over layers at α = 1.0.



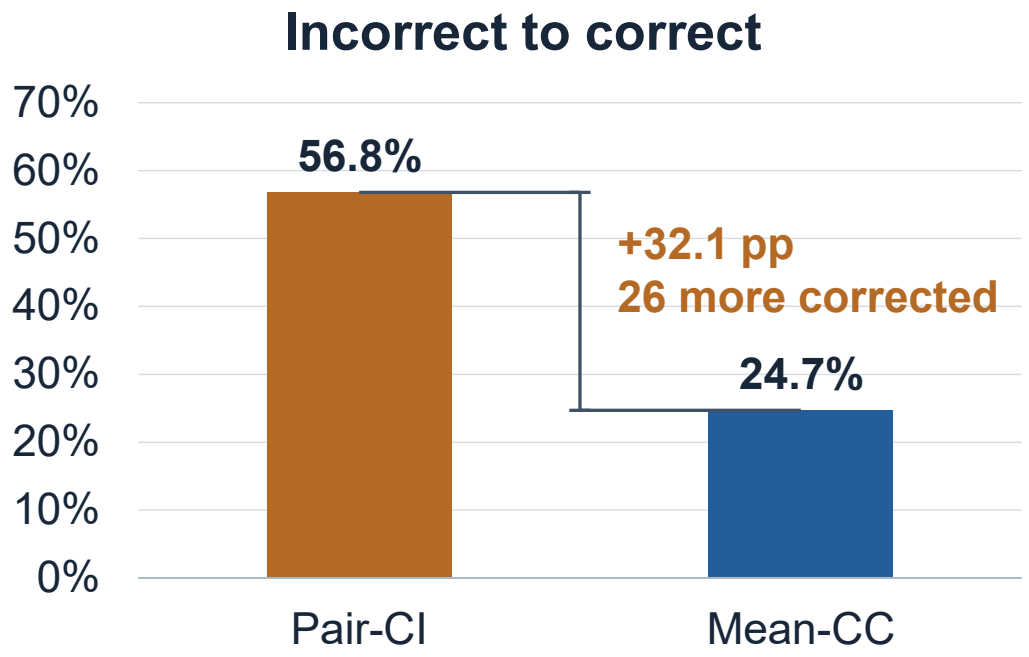

**The tested group and lexical controls do not reproduce the Pair-CI peak.**

# Correcting errors with little disruption

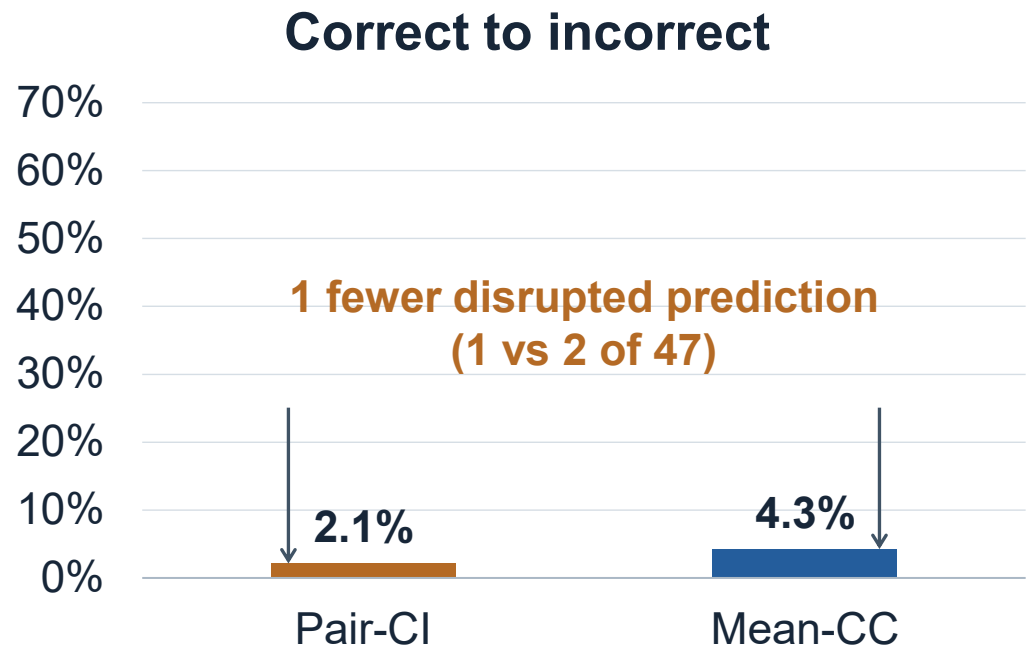


Gemma 4 E4B IT, layer 28, α = 1.0

81 initially incorrect predictions




47 initially correct predictions




**At this layer, Pair-CI corrects more errors with little disruption.**

# Limitations



| | |
|---|---|
| **Meaning** | Semantic content and full solution structure remain unidentified. |
| **Scope** | One dataset, four models, one failure group.<br>One prompt position and one layer at a time. |
| **Peak estimates** | Descriptive layer maxima.<br>No held-out selection or uncertainty estimates. |

# Answer to the research question



Does a **problem-specific hidden state** recover reasoning failures more effectively than an **averaged hidden state**?

## **Problem-specific states** give higher peak recovery.

All four tested models, α = 1.0 (Pair-CI vs Mean-CC)

Pair-CI on E4B, layer 28: more error correction with little disruption of correct predictions.

**Next** Identify the useful information and test broader reasoning tasks.